

ADITYA TANWAR

Junior Undergraduate

Department of Computer Science and Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/ Certificate	Institute	CPI/%
2024 (expected)	B.Tech	Indian Institute of Technology Kanpur	9.9/10
2020	CBSE (XII)	Delhi Public School	94.8%
2018	CBSE (X)	Delhi Public School	87.0%

SCHOLASTIC ACHIEVEMENTS

- Received **A*** grade in 8 courses for outstanding performance at IIT-Kanpur (2020-Present)
- Received **Academic Excellence Award** twice for exceptional performance at IIT-Kanpur (2020, '21)
- Course **topper** of **ESO207 - Data Structures & Algorithms** in a class of **251** students at IIT-Kanpur (2021)
- Received the **Class of 1990 Scholarship** for being amongst the **top-3 JEE (Advanced) rankers** at IIT-Kanpur (2020)
- Secured **All India Rank 119** in **JEE-Advanced** among 240,000+ eligible candidates (2020)
- Secured **All India Rank 81** with **100th percentile** in **Mathematics in JEE-Main** among 1,000,000+ candidates (2020)
- Recipient of the prestigious **KVPY Fellowship** by DST, Govt. of India **twice** in SA and SX stream (2018, '19)

PROJECTS

- SAT solver** 🌀 (Course Project: Logic for Computer Science, Prof. Subhajit Roy) (Jan'22-Feb'22)
 - * **Encoded** the problem of Pair Sudoku solving into that of a **boolean satisfiability** and used **PySAT** to find solutions
 - * Explored and verified the **qualitative effect of redundancy** in encoding and constraints on runtime of SAT solvers
 - * Studied and employed the **DPLL algorithm** with **branching heuristics (MOMS)** in C++ to design a SAT solver
- Blocks and Chains** 🌀 (Association for Computing Activities, IIT-Kanpur) (Apr'21-Jul'21)
 - * Studied the foundations of **blockchains** along with pertinent **cryptographic concepts** like **hashing** and **encryption**
 - * Implemented **signature verification, transaction data reader**, etc. in **Javascript** using libraries like **Crypto**
- Model Zoo** (Science and Technology Council, IIT-Kanpur) (May'21-Jul'21)
 - * Constructed a **CNN model** to predict a number (0-9), given an audio file of its pronunciation as a pre-requisite 🌀
 - * Grasped standard techniques in **deep learning**, specifically those used in a Convolutional Neural Network (CNN)
 - * Implemented state of the art deep learning research papers like **Imagenet, ResNet, RepVGG** in **TensorFlow** 🌀

PROGRAMMING ACHIEVEMENTS

- Competent in programming with maximum ratings: **1803 (expert)** on **Codeforces**, **2168 (5*)** on **Codechef** (2020-Present)
- Global rank **304** in **Google Kick Start – Round D**, **22** in **Codechef Starters Div. 2**, **194** in **Codeforces Round #788** (2022)
- Won** the November edition of **Sport Programmer of the Month** organized by Programming Club, IIT-Kanpur (2021)

RELEVANT COURSEWORK

Data Structures & Algorithms (*) Linear Algebra & ODE (*) Probability for Computer Science Logic for Computer Science	Fundamentals of Computing Discrete Mathematics Software Development and Operations Complex Analysis (o)	Real Analysis (*) Introduction to Electronics (*) Computer Organization Macroeconomics (o)
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(*): A* Grade for outstanding performance, (o): ongoing

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python, Javascript, Verilog, MIPS
- Libraries & Utilities:** Git, $\LaTeX$, Bash, TensorFlow, Keras, Matplotlib, PySAT

POSITION OF RESPONSIBILITY

- Project Mentor, Under The Hood** (Programming Club, IIT-Kanpur) (May'22-Jul'22)
 - * **Prepared** and **delivered lectures** to a class with **more than 20** freshmen and sophomores, and responded to queries
 - * Explained topics pertaining to **boolean algebra**, basic **electronics**, **HDLs** and **low level programming** languages
 - * **Created problem statements** for assignments and **organised** mentees into groups to work on a project together
- Academic Mentor, ESC101** (Counselling Service, IIT-Kanpur) (Dec'21-Jul'22)
 - * Entrusted to help freshmen with their academics in general and **ESC101 - Fundamentals of Computing** in particular
 - * **Resolved difficulties** and **illustrated** several general **programming paradigms** in the **C Programming Language**
 - * **Mentored** academically weak students one-to-one and delivered institute level **remedial classes** for **150+** freshers
- Secretary** (Programming Club, IIT-Kanpur) (Aug'21-May'22)
 - * Applied a modified version of **Gale-Shapley Algorithm** for the allotment of summer projects to **1000+** students
 - * Formulated **multiple problem statements** (competitive programming) for the **Fresher's Programming Contest**
 - * **Managed** the September edition of "Sport Programmer of the Month", a competitive programming event